Q29 - Transfer Learning in Deep Neural Networks for Identifying Safe Drinking Water Sources (SDG 6)

Introduction

Transfer learning is a machine learning technique in which a deep neural network trained on a large dataset is reused and fine-tuned for a related task. Instead of training a model from scratch, previously learned knowledge is transferred to a new problem. In rural regions, where only a limited amount of water quality data is available, transfer learning helps develop accurate models for identifying safe and unsafe drinking water sources.

Need for Transfer Learning

Collecting large amounts of labeled water quality data in rural areas is expensive and time-consuming. Limited laboratory facilities, seasonal variations, and shortage of experts make data collection difficult. Transfer learning reduces the need for massive datasets while improving prediction accuracy.

Working

A pre-trained model such as ResNet, MobileNet, or EfficientNet is selected. The early layers, which have already learned useful features, are retained. The final classification layer is replaced with a new layer that predicts whether water is Safe or Unsafe. The model is then fine-tuned using a small local dataset containing features such as pH, turbidity, dissolved oxygen (DO), total dissolved solids (TDS), temperature, and conductivity.

Advantages

Transfer learning requires less training data, reduces computational cost, speeds up training, improves prediction accuracy, minimizes overfitting, and is ideal for rural water monitoring applications where labeled data is scarce.

Applications

Applications include groundwater quality monitoring, river and lake pollution detection, IoT-based smart water monitoring systems, contamination alerts, and rural drinking water assessment.

Contribution to SDG 6

Transfer learning supports SDG 6 (Clean Water and Sanitation) by enabling affordable and reliable monitoring of water quality, improving access to safe drinking water, detecting contamination early, and supporting sustainable management of water resources.

Conclusion

Transfer learning is a practical solution for building accurate deep learning models in data-scarce rural regions. By reusing knowledge from pre-trained networks, it provides faster training, better accuracy, and lower computational cost, making it highly suitable for identifying safe drinking water sources and promoting sustainable development.